



8/7/2026

# ASSIGNMENT DAY5

Network Setup & Connectivity Testing.

## TASK 1.

- ❖ Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.



```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\callm>ping 192.168.1.150

Pinging 192.168.1.150 with 32 bytes of data:
Reply from 192.168.1.150: bytes=32 time=874ms TTL=64
Reply from 192.168.1.150: bytes=32 time=91ms TTL=64
Reply from 192.168.1.150: bytes=32 time=89ms TTL=64
Reply from 192.168.1.150: bytes=32 time=97ms TTL=64

Ping statistics for 192.168.1.150:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 89ms, Maximum = 874ms, Average = 287ms

C:\Users\callm>
```

Figure : Successful ping test verifying communication between the laptop and smartphone.

### Explanation :

- ✓ The laptop and a friend's phone were connected to the same Wi-Fi network. A ping test was performed from the laptop using the phone's IP address to verify network connectivity. The test completed successfully with 4 packets sent, 4 packets received, and 0% packet loss, confirming successful communication between both devices on the same network.

## TASK 2.

- ❖ Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to [www.youtube.com](http://www.youtube.com) from your device and note down the number of hops it takes to reach the destination.



```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\callm>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482b:400::]
over a maximum of 30 hops:

  1    27 ms    11 ms    2 ms  2401:4900:8fec:b84b:6e4f:89ff:fef3:dc61
  2      *      *      *    Request timed out.
  3     6 ms     4 ms     5 ms  fc00::93
  4    10 ms    39 ms     4 ms  2404:a800:2a00:103::2d
  5    31 ms    43 ms    25 ms  2404:a800::167
  6    36 ms    43 ms    26 ms  2001:4860:482b:400::

Trace complete.

C:\Users\callm>
```

Figure : Traceroute result showing the network path to [www.youtube.com](http://www.youtube.com).

### Explanation :

- ✓ The tracert command was executed from the Windows Command Prompt to trace the network path to [www.youtube.com](http://www.youtube.com). The destination was reached successfully in 6 hops. This shows the route taken by the data packets from the device to the destination.

 **Number of Hops: 6**

## TASK 3.

- ❖ Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.



```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\callm>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482b:400::]
over a maximum of 30 hops:

  1  27 ms  11 ms  2 ms  2401:4900:8fec:b84b:6e4f:89ff:fef3:dc61
  2  *      *      *      Request timed out.
  3  6 ms   4 ms   5 ms   fc00::93
  4  10 ms  39 ms  4 ms   2404:a800:2a00:103::2d
  5  31 ms  43 ms  25 ms  2404:a800::167
  6  36 ms  43 ms  26 ms  2001:4860:482b:400::

Trace complete.

C:\Users\callm>
```

Figure : Ping test after disconnecting the device from the Wi-Fi network.

### Explanation :

- ✓ One device was disconnected from the Wi-Fi network, and the ping command was executed again from the laptop. The ping test failed, displaying the error message “Destination host unreachable.” This indicates that the disconnected device could no longer be reached over the network.

 **Error Message: Destination host unreachable.**

# TASK 4.

- ❖ Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.



```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\callm>arp -a

Interface: 192.168.1.169 --- 0x5
Internet Address      Physical Address      Type
192.168.1.1           6c-4f-89-f3-dc-61     dynamic
192.168.1.30          c4-75-ab-e7-a5-ea     dynamic
192.168.1.33          30-03-c8-ac-55-f3     dynamic
192.168.1.45          98-fe-3e-5f-18-2e     dynamic
192.168.1.52          f4-7b-09-27-0f-17     dynamic
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.148.1 --- 0xd
Internet Address      Physical Address      Type
192.168.148.254       00-50-56-fd-29-cd     dynamic
192.168.148.255       ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.24.1 --- 0x14
Internet Address      Physical Address      Type
192.168.24.254        00-50-56-ff-c8-db     dynamic
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\callm>
```

Figure : ARP table showing the IP addresses of devices connected to the WI FI network.

## Explanation :

- ✓ The arp -a command was executed in the Windows Command Prompt to display the IP addresses of devices connected to the Wi-Fi network. The detected devices were identified based on the available network information and are listed in the table below.

## Connected Devices :

| IP Address    | Device / Person     |
|---------------|---------------------|
| 192.168.1.1   | WI FI Router        |
| 192.168.1.169 | ASUS TUF F15 Laptop |
| 192.168.1.150 | My Iphone           |

## TASK 5.

- ❖ **Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Include screenshots or command outputs if possible.**

### ➤ **Network Setup, Connectivity Testing, and Observations :**

#### **Introduction :**

- ✓ I connected my laptop to the home Wi-Fi network and confirmed that the internet connection was active. After the connection was established, I opened Command Prompt to test the network connectivity and verify that the connection was working properly.

#### **Step 1 - Ping Test :**

- ✓ First, I used the command ping google.com to check whether my device could communicate with the internet. The command returned successful replies with normal response times, confirming that the network connection was active and stable.

➤ Figure 1 is presented on the following page due to page layout constraints.

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\callm>tracert google.com

Tracing route to google.com [142.250.134.138]
over a maximum of 30 hops:

  1    1 ms    1 ms    1 ms  192.168.167.1
  2    3 ms    1 ms    1 ms  192.168.5.1
  3    3 ms    2 ms    2 ms  202.131.108.233
  4    5 ms    2 ms    3 ms  180.211.104.2
  5    9 ms    2 ms    5 ms  202.131.101.133
  6    5 ms    2 ms    3 ms  202.131.98.241
  7    4 ms    2 ms    2 ms  202.131.108.13
  8   13 ms   12 ms   11 ms  202.131.99.94
  9   12 ms   11 ms   12 ms  27.109.1.197
 10   14 ms   11 ms   12 ms  202.131.99.201
 11   13 ms   13 ms   12 ms  172.253.69.227
 12   30 ms   11 ms   11 ms  192.178.110.208
 13   14 ms   12 ms   12 ms  108.170.236.159
 14   21 ms   20 ms   36 ms  172.253.76.210
 15    *      *      *    Request timed out.
 16    *      *      *    Request timed out.
 17    *      *      *    Request timed out.
 18    *      *      *    Request timed out.
 19    *      *      *    Request timed out.
 20    *      *      *    Request timed out.
 21   15 ms   13 ms   14 ms  fx-in-f138.1e100.net [142.250.134.138]

Trace complete.

C:\Users\callm>|
```

Figure 1 : Ping Command Output.

## Step 2 - Traceroute Test :

- ✓ Next, I used the command `tracert google.com` to trace the path taken by data packets from my computer to Google's server. The output displayed multiple network hops, showing how the packets traveled through different routers before reaching their destination.

➞ Figure 2 is presented on the following page due to page layout constraints.

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\callm>ping google.com

Pinging google.com [192.178.174.138] with 32 bytes of data:
Reply from 192.178.174.138: bytes=32 time=27ms TTL=109
Reply from 192.178.174.138: bytes=32 time=14ms TTL=109
Reply from 192.178.174.138: bytes=32 time=14ms TTL=109
Reply from 192.178.174.138: bytes=32 time=14ms TTL=109

Ping statistics for 192.178.174.138:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 27ms, Average = 17ms

C:\Users\callm>|
```

Figure 2 : Traceroute (Tracert) Command Output.

## Observations

- ☞ The laptop connected to the Wi-Fi network successfully.
- ☞ The ping test received successful replies, confirming that the internet connection was working correctly.
- ☞ The traceroute command displayed the complete path of the data packets through multiple network hops.
- ☞ No major connectivity issues or network errors were observed during the testing.

## Conclusion

- ✓ The network was set up successfully, and both connectivity tests confirmed that the internet connection was functioning properly. The ping command verified network connectivity, while the traceroute command showed the route taken by data packets. These tests helped in understanding the network performance and the communication process over the internet.